A^3-ENDS

Weekly Security Operations Center (SOC) Report

Report Number: WK-20260708

Reporting Period: 2026-07-01 to 2026-07-08

Organization: Enterprise Network

Generated By: A3-ENDS Automated Pipeline

Date Generated: 2026-07-08 19:35:57 UTC

Confidentiality Level: RESTRICTED

Digital Verification: VALID

1. Executive Summary & 2. Weekly Security Dashboard

Reporting Period: 2026-07-01 - 2026-07-08

EXECUTIVE SUMMARY

This week, the SOC analyzed 6,561 total network flows, identifying 19 confirmed attacks with 5 false positives, reflecting a high accuracy rate in threat detection. The severity breakdown highlights 14 HIGH, 4 MEDIUM, and 1 LOW severity incidents, with no CRITICAL or INFO-level alerts. The threat landscape was dominated by ZERO_DAY_ANOMALY attacks (14 instances, highest risk score of 84.99) and DOS attacks (5 instances, highest risk score of 59.99). Automated responses effectively mitigated these threats, ensuring minimal operational disruption. Continued vigilance is recommended, particularly for zero-day anomalies, which pose the highest risk. The SOC remains proactive in refining detection mechanisms to reduce false positives and enhance response efficiency.

TOTAL NETWORK FLOWS	PACKETS CAPTURED	BENIGN FLOWS	SUSPICIOUS FLOWS
6561	98415	261	6300

CONFIRMED ATTACKS	FALSE POSITIVES	AVERAGE RISK SCORE	HIGHEST RISK SCORE
19	5	76/100	84/100

ANALYST DECISIONS (APPROVED)	ANALYST DECISIONS (REJECTED)	AVERAGE ANALYST RESPONSE TIME
11	5	245s

3. Attack Overview & 5. Severity Distribution

Attack Overview

ATTACK TYPE	COUNT	HIGHEST RISK
DOS	5	59
ZERO_DAY_ANOMALY	14	84

Severity Distribution

SEVERITY	COUNT
Critical	0
High	14
Medium	4
Low	1

7. AI Detection Performance (SHAP Feature Importance)

TOP FEATURES INFLUENCING AI DECISIONS THIS WEEK

FEATURE NAME	AGGREGATED SHAP IMPACT
mean_packet_size	Score: 32.97
packet_count	Score: 26.06
std_packet_size	Score: 22.77
mean_iat	Score: 15.60
duration	Score: 9.62

total_bytes

Score: 5.85

10. Concept Drift Detection & 11. Top IOCs

ADWIN DRIFT EVENTS	Q-LEARNING UPDATES	TOTAL POLICY UPDATES
2	38	40

Top Indicators of Compromise (Highest Risk Alerts)

SOURCE IP	DESTINATION IP	PORT	PROTOCOL	ATTACK	RISK	STATUS
192.168.56.1	192.168.56.101	12345	17	**ZERO_DAY_ANOMALY**	84	NEW
192.168.56.1	192.168.56.101	12345	17	**ZERO_DAY_ANOMALY**	84	NEW
192.168.56.1	192.168.56.101	12345	17	**ZERO_DAY_ANOMALY**	84	NEW
192.168.56.1	192.168.56.101	12345	17	**ZERO_DAY_ANOMALY**	84	NEW
192.168.56.101	192.168.56.1	0	6	**ZERO_DAY_ANOMALY**	84	NEW

12. Top Talkers

TOP SOURCE IPS	COUNT
192.168.56.1	13
192.168.1.206	1
192.168.1.248	1
192.168.56.101	1
192.168.1.201	1

TOP DESTINATION IPS	COUNT
192.168.56.101	12
192.168.56.1	6
239.255.255.250	1

13. Network Health & 16. System Status

AVERAGE THROUGHPUT

1.2 Gbps

AVERAGE LATENCY

14 ms

SYSTEM STATUS

- Packet Capture
- Autoencoder (Anomaly)
- LightGBM (Signature)
- SHAP Explainer
- Risk Engine
- SQLite Database
- LLM Reporting
- ADWIN Drift Detector

15. Security Recommendations

Given the prevalence of Zero-Day Anomaly attacks and the recurring targeting of port 12345, it is critical to implement immediate firewall tuning to restrict access to this port and enforce stricter ingress/egress rules to mitigate potential exploitation. Additionally, deploying Multi-Factor Authentication (MFA) across all critical systems can significantly reduce the risk of unauthorized access, particularly in light of the high-risk Zero-Day Anomaly incidents. Finally, consider architectural changes such as segmenting the network to limit lateral movement and deploying advanced threat detection mechanisms to enhance visibility and response capabilities against sophisticated attacks like DoS and Zero-Day exploits.

17. Appendix & 18. Analyst Signatures

Model Version: v1.2 (Retrained CICIDS2017)

LightGBM Version: 4.3.0

TensorFlow Version: 2.15.0

Report Hash (SHA256): 8f4e3c1b... (Verified)

Prepared By: ______________________________

Reviewed By: ______________________________

Date: 2026-07-08 19:35:57 UTC